

Q. What is Regularization?

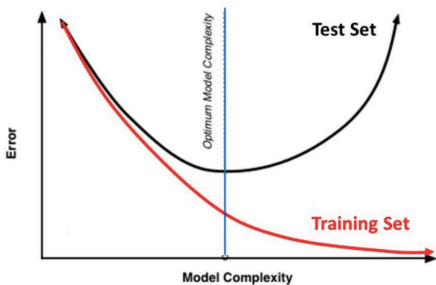
→ Regularization is a technique used in machine learning and deep learning to prevent overfitting and improve model's generalization performance. It involves adding a penalty term to the loss function during training.

This penalty discourages the model from becoming too complex or having large parameter values, which helps in controlling the model's ability to fit noise in the training data.

Regularization in deep learning methods includes

- 1) L1 and L2 regularization
- 2) dropout
- 3) early stopping

Training Vs. Test Set Error



Ridge and Lasso Regression :

Here's a table outlining the key differences between ridge vs lasso regression:

Feature	Ridge Regression	Lasso Regression
Description	Ridge regression, also known as Tikhonov regularization, is a technique that introduces a penalty term to the linear regression model to shrink the coefficient values.	Lasso regression, or Least Absolute Shrinkage and Selection Operator, is a regularization method that also includes a penalty term but can set some coefficients exactly to zero, effectively selecting relevant features.
Penalty Type	Ridge regression utilizes an L2 penalty, which adds the sum of the squared coefficient values multiplied by a tuning parameter (λ).	Lasso regression employs an L1 penalty, which sums the absolute values of the coefficients multiplied by λ .
Coefficient Impact	The L2 penalty in ridge regression discourages large coefficient values, pushing them towards zero but never exactly reaching zero. This shrinks the less important features' impact.	The L1 penalty in lasso regression can drive some coefficients to exactly zero when the λ value is large enough, performing feature selection and resulting in a sparse model.
Feature Selection	Ridge regression retains all features in the model, reducing the impact of less important features by shrinking their coefficients.	Lasso regression can set some coefficients to zero, effectively selecting the most relevant features and improving model interpretability.
Use Case	Ridge regression is useful when the goal is to minimize the impact of less important features while keeping all variables in the model.	Lasso regression is preferred when the goal is feature selection, resulting in a simpler and more interpretable model with fewer variables.
Model Complexity	Ridge regression tends to favor a model with a higher number of parameters, as it shrinks less important coefficients but keeps them in the model.	Lasso regression can lead to a less complex model by setting some coefficients to zero, reducing the number of effective parameters.
Interpretability	The results of ridge regression may be less interpretable due to the inclusion of all features, each with a reduced but non-zero coefficient.	Lasso regression can improve interpretability by selecting only the most relevant features, making the model's predictions more explainable.
Sparsity	Ridge regression does not yield sparse models since all coefficients remain non-zero.	Lasso regression can produce sparse models by setting some coefficients to exactly zero.
Sensitivity	More robust and less sensitive to outliers compared to lasso regression.	More sensitive to outliers due to the absolute value in the penalty term.

	rss	intercept	coef_x_1	coef_x_2	coef_x_3	coef_x_4	coef_x_5	coef_x_6	coef_x_7	coef_x_8	coef_x_9	coef_x_10	coef_x_11	c
model_pow_1	3.3	2	-0.62	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	h
model_pow_2	3.3	-1.9	-0.58	-0.006	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	h
model_pow_3	1.1	-1.1	3	-1.3	0.14	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	h
model_pow_4	1.1	-0.27	1.7	-0.53	-0.036	0.014	NaN	NaN	NaN	NaN	NaN	NaN	NaN	h
model_pow_5	1	3	-5.1	4.7	-1.9	0.33	-0.021	NaN	NaN	NaN	NaN	NaN	NaN	h
model_pow_6	0.99	-2.8	6.5	-9.7	5.2	-1.6	0.23	-0.014	NaN	NaN	NaN	NaN	NaN	h
model_pow_7	0.93	19	-58	69	-45	17	-3.5	0.4	-0.019	NaN	NaN	NaN	NaN	h
model_pow_8	0.92	43	-1.4e+02	1.8e+02	-1.3e+02	58	-15	6.4	-0.21	0.0077	NaN	NaN	NaN	h
model_pow_9	0.87	1.7e+02	-6.1e+02	9.6e+02	-8.5e+02	4.6e+02	-1.6e+02	37	-5.2	0.42	-0.015	NaN	NaN	h
model_pow_10	0.87	1.4e+02	-4.9e+02	7.3e+02	-6e+02	2.9e+02	-87	15	-0.81	-0.14	0.026	-0.0013	NaN	h
model_pow_11	0.87	-75	6.1e+02	-1.3e+03	1.9e+03	-1.6e+03	9.1e+02	-3.5e+02	91	-16	1.8	-0.12	0.0034	h
model_pow_12	0.87	-3.4e+02	1.3e+03	-4.4e+03	6e+03	-5.2e+03	3.1e+03	-1.3e+03	3.8e+02	-80	12	-1.1	0.062	-i
model_pow_13	0.86	3.2e+03	-1.8e+04	4.5e+04	-6.7e+04	6.6e+04	-4.6e+04	2.3e+04	-8.5e+03	2.3e+03	-4.5e+02	62	-5.7	0
model_pow_14	0.79	2.4e+04	-1.4e+05	3.8e+05	-6.1e+05	6.6e+05	-5e+05	2.6e+05	-1.2e+05	3.7e+04	-8.5e+03	1.5e+03	-1.8e+02	1
model_pow_15	0.7	-3.6e+04	2.4e+05	-7.5e+05	1.4e+06	-1.7e+06	1.5e+06	-1e+06	5e+05	-1.9e+05	5.4e+04	-1.2e+04	1.9e+03	-i

Note: size of coefficients increases exponentially with increase in model complexity.
This gives intuition into why putting a constraint on the magnitude of coefficients can be good idea to reduce model complexity.

Ridge Regression:-
(L2 Regularization)

$RSS + \alpha * (\text{sum of square of coefficients})$

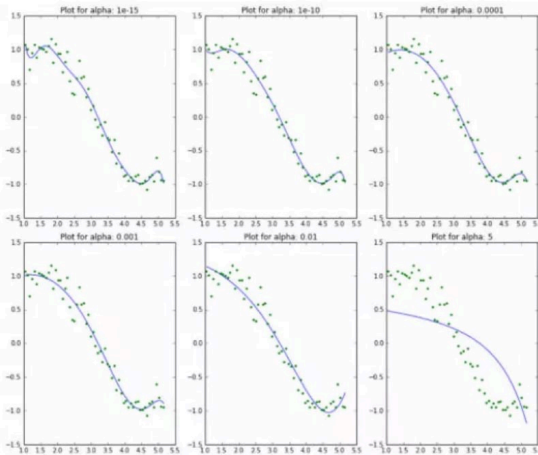
α : It is the parameter that balances the amount of emphasis given to minimizing RSS vs minimizing the sum of squares of coefficients.

$\alpha = 0$: The objective becomes same as loss function of linear regression.

$\alpha = \infty$: The coefficients will be zero because. Because of infinite weightage on square of coefficients, anything less than zero will make objective function ∞ .

Note: As the value of alpha increases, the model complexity

reduces. Though higher value of alpha reduces overfitting, significantly high values can cause underfitting as well.



	rss	intercept	coef_x_1	coef_x_2	coef_x_3	coef_x_4	coef_x_5	coef_x_6	coef_x_7	coef_x_8	coef_x_9	coef_x_10	coef_x_11	coef
alpha_1e-15	0.87	95	-3e+02	3.8e+02	-2.4e+02	66	0.96	-4.8	0.64	0.15	-0.026	-0.0054	0.00086	0.0
alpha_1e-10	0.92	11	-29	31	-15	2.9	0.17	-0.091	-0.011	0.002	0.00064	2.4e-05	-2e-05	-4.1
alpha_1e-08	0.95	1.3	-1.5	1.7	-0.68	0.039	0.016	0.00016	-0.00036	-5.4e-05	-2.9e-07	1.1e-06	1.9e-07	2e-
alpha_0.0001	0.96	0.56	0.55	-0.73	-0.026	-0.0028	-0.00011	4.1e-05	1.5e-05	3.7e-06	7.4e-07	1.3e-07	1.9e-08	1.9
alpha_0.001	1	0.82	0.31	-0.067	-0.02	-0.0028	-0.00022	1.8e-05	1.2e-05	3.4e-06	7.3e-07	1.3e-07	1.9e-08	1.7
alpha_0.01	1.4	1.3	-0.088	-0.052	-0.01	-0.0014	-0.00013	7.2e-07	4.1e-06	1.3e-06	3e-07	5.6e-08	9e-09	1.1
alpha_1	5.6	0.97	-0.14	-0.019	-0.003	-0.00047	-7e-05	-9.9e-05	-1.3e-05	-1.4e-07	-5.3e-09	1.3e-09	7.8e-10	2.4
alpha_5	14	0.55	-0.059	-0.0085	-0.0014	-0.00024	-4.1e-05	-6.9e-05	-1.1e-05	-1.9e-07	-5.1e-08	-5.1e-09	-8.2e-10	-1.1
alpha_10	18	0.4	-0.037	-0.0055	-0.00095	-0.00017	-3e-05	-5.2e-05	-9.2e-07	-1.6e-07	-2.8e-08	-5.1e-09	-9.1e-10	-1.1
alpha_20	23	0.28	-0.022	-0.0034	-0.0006	-0.00011	-2e-05	-3.6e-05	-6.6e-07	-1.2e-07	-2.2e-08	-4e-09	-7.5e-10	-1.1

Lasso Regression : $RSS + \alpha * (\text{sum of absolute values of coefficients})$

	rss	intercept	coef_x_1	coef_x_2	coef_x_3	coef_x_4	coef_x_5	coef_x_6	coef_x_7	coef_x_8	coef_x_9	coef_x_10	coef_x_11	coef
alpha_1e-15	0.96	0.22	1.1	-0.37	0.00089	0.0016	-0.00012	-6.4e-05	-6.3e-06	1.4e-06	7.8e-07	2.1e-07	4e-08	5.4
alpha_1e-10	0.96	0.22	1.1	-0.37	0.00088	0.0016	-0.00012	-6.4e-05	-6.3e-06	1.4e-06	7.8e-07	2.1e-07	4e-08	5.4
alpha_1e-08	0.96	0.22	1.1	-0.37	0.00077	0.0016	-0.00011	-6.4e-05	-6.3e-06	1.4e-06	7.8e-07	2.1e-07	4e-08	5.3
alpha_1e-05	0.96	0.5	0.6	-0.13	-0.038	0	0	0	0	7.7e-06	1e-06	7.7e-08	0	0
alpha_0.0001	1	0.9	0.17	-0	-0.048	-0	-0	0	0	9.5e-06	5.1e-07	0	0	0
alpha_0.001	1.7	1.3	-0	-0.13	-0	-0	-0	0	0	0	0	0	1.5e-08	7.5
alpha_0.01	3.6	1.8	-0.55	-0.00056	-0	-0	-0	-0	-0	-0	-0	-0	0	0
alpha_1	37	0.038	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	0
alpha_5	37	0.038	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	0
alpha_10	37	0.038	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	-0	0

HIGH SPARSITY

x : the matrix of input features (nrow: N , ncol: $M+1$)

y : the actual outcome variable (length: N)

$\hat{y}$: predicted values of Y (length: N)

w : the weights on the coefficients (length: $M+1$)

$$\hat{y}_i = \sum_{j=0}^M w_j * x_{ij}$$

Simple Linear Regression Model:-

$$\text{Cost}(w) = \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \sum_{i=1}^N \left(y_i - \sum_{j=0}^M w_j x_{ij} \right)^2$$

$$\frac{\partial}{\partial w_j} \text{Cost}(w) = -2 \sum_{i=1}^N x_{ij} \left\{ y_i - \sum_{j=0}^M w_j x_{ij} \right\}$$

$\therefore$ By Gradient descent

$$w_j^{t+1} = w_j^t + 2\eta \sum_{i=1}^N x_{ij} \left\{ y_i - \sum_{k=0}^M w_k x_{ik} \right\}$$

Ridge Regression :

$$\begin{aligned} \text{Cost}(w) &= \text{RSS}(w) + \lambda \sum_{j=0}^M w_j^2 \\ &= \sum_{i=1}^N \left\{ y_i - \sum_{j=0}^M w_j x_{ij} \right\}^2 + \lambda \sum_{j=0}^M w_j^2 \end{aligned}$$

$$\frac{\partial}{\partial w_j} \text{Cost}(w) = -2 \sum_{i=1}^N x_{ij} \left\{ y_i - \sum_{k=0}^M w_k x_{ik} \right\} + 2\lambda w_j$$

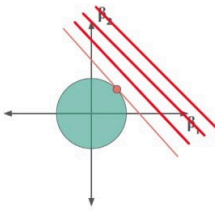
$$w_j^{t+1} = (1 - 2\lambda\eta) w_j^t + 2\eta \sum_{i=1}^N m_{ij} \left\{ y_i - \sum_{k=0}^M w_k x_{ik} \right\}$$

Note:- Ridge regression is equivalent to reducing the weight by a factor of $(1 - 2\lambda\eta)$ first and then applying the same update rule as linear regression.

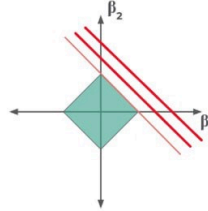
Lasso Regression:-

$$\text{Cost}(w) = \text{RSS}(w) + \lambda \sum_{j=0}^M |w_j|$$

- Penalty for Ridge: $\beta_1^2 + \beta_2^2 \leq s$



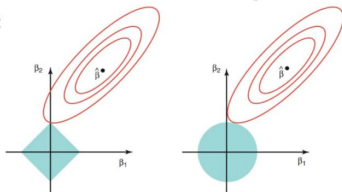
- Penalty for Lasso: $|\beta_1| + |\beta_2| \leq s$



- Lasso:

- A convex object that lies tangent to the boundary, is likely to encounter a corner of a hypercube, for which some components of β are identically zero.

- This is why Lasso is more likely to lead to coefficients as zero.
- This diagram is also commonly shown with contour RSS:



Elastic Net:-

$$L = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda_1 \sum_{j=1}^p \beta_j^2 + \lambda_2 \sum_{j=1}^p |\beta_j|$$

alternatively we can express this as ratio between L_1 and L_2 :

$$\frac{\sum_{j=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ij})^2}{2n} + \lambda \left(\frac{1-\alpha}{2} \sum_{j=1}^n \beta_j^2 + \alpha \sum_{j=1}^n |\beta_j| \right)$$



